K Prakash

B.Tech Biotechnology and Bioinformatics

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Degree	University/Institute	Year	CGPA/Marks(%)
B.Tech Biotechnology and Bioinformatics	IIT Hyderabad	2027	8.07
Minor in Economics	IIT Hyderabad	2027	N.A
XII (TSBIE)	TSWRS BELLAMPALLY	2022	9.57
X (BSE(Telangana))	TSWRS CHILKOOR	2020	9.80

POSITIONS OF RESPONSIBILITY

Current Position 2025-26

Coordinator**,** Office of Career Services, IIT Hyderabad.

- Coordinating placement and internship activities; facilitating communication between students and recruiters.

Past Experience 2024-25

Coordinator, **Design & PR, Entrepreneurship** and Makers Lab, IIT Hyderabad.

- Led design and public relations initiatives.
- Managed digital content, branding, and promotional strategy.

SKILLS

Programming: C/C++, Python, R and SQL
Web Development: HTML, CSS, JavaScript, React.js, Node.js, and MySQL.
Software Tools : Git, GitHub, LaTeX, and MS Excel
Libraries/Frameworks: NumPy, Pandas, Matplotlib, SciPy, Seaborn.
OS: Linux and Windows.

RELEVANT COURSES

Computer Science: Introduction to programming, R, and Python for Biologists.
AI and ML: Introduction to Artificial Intelligence, Big Data Biology and Biological Databases, Foundation of Machine Learning and Bio-Statistics.
Mathematics: Probability and Random Variables, Calculus I, Calculus II, Differential Equations, Linear Algebra.

PROJECTS

Predicting Coronavirus Outbreak Using Machine Learning

- Designed and implemented machine learning models (**Polynomial Regression**, **Support Vector Regression**) to forecast COVID-19 outbreak trends using global time-series data.
- Preprocessed large epidemiological datasets, handling missing data, feature scaling, and training/testing split to build robust predictive models.
- Evaluated and compared model performance to identify the best approach for capturing non-linear pandemic growth patterns.
- Analyzed key factors such as days since the first confirmed case to understand outbreak dynamics and improve model insights.
- Explored enhancements using advanced feature engineering and deep learning techniques to boost predictive accuracy.

Clinic Management System (C++ Project)

- Created and applied a C++-based patient management system with emergency queue priority through the application of OOP principles and STL data structures.
- Provided patient registration with fields such as name, age, gender, blood group, and medical issue; stored records automatically in a CSV file for persistence.
- Developed functionality for emergency triaging, display of queues, patient search by mobile number, and assignment of doctors.
- Successfully emulated real-time emergency management and patient flow in a clinical environment using file I/O, maps, and queues.

Customer Segmentation – Machine Learning Project

Python · Pandas · Scikit-learn · Matplotlib · Seaborn

- Designed and implemented an **end-to-end machine learning pipeline** for customer segmentation on **2,200+ records** using **K-Means clustering** and **t-SNE** visualization.
- Conducted **data pre-processing, feature scaling, and normalization** to improve clustering performance.
- Applied the **Elbow Method** to identify the optimal number of clusters, revealing **6 meaningful customer groups**.
- Created **data-driven visualizations** (heatmaps, scatter plots) to interpret cluster characteristics and support business insights.

EXTRACURRICULAR

- NSO Cricket player, runner-up in Institute Cricket League(ICL). 2023
- Proud **NCC cadet**; won **1st prize in parade.**